Engineering Evaluation: Technical Feasibility, Silicon Revision Forensics, OTP Decode Matrix, and Production Configurations for 1.54-Inch (200×200) Electrophoretic Displays

Empirical Assessment of Solomon Systech SSD1681, HINK-E0154A07-A1 (Lot SYX 1920), GDEH0154D67 / GDEY0154D67, and Waveshare 1.54" 200×200 E-Paper Variants

Metadata Attribute	Engineering Specification & Determination
Evaluation Status	**2-Bit Grayscale Restricted to Laboratory Evaluation; 1-Bit Bi-Tonal Full & Partial Refresh Approved for Production**
Architecture Domain	Display Driver Firmware, OTP Register Forensics, Electro-Optical Physics & Embedded Typography Subsystem
Target Hardware	Solomon Systech SSD1681 COG Driver / `HINK-E0154A07-A1` (FPC Date `2017-02-28`, Lot `SYX 1920`) / Good Display GDEH0154D67 & GDEY0154D67 / Waveshare 1.54" 200×200 Bi-Tonal E-Paper
Primary Objective	Establish a deterministic OTP hexadecimal reference matrix, waveform configuration specification, production-readiness evaluation, and reference C implementation for 1-bit full/partial refresh and experimental 2-bit grayscale
Document Date & Rev	November 5, 2026 — Revision 3.0 (Final Formal Engineering Specification)

1. Executive Summary and Engineering Determination

Electrophoretic displays (EPDs) operate on microencapsulated electrophoretic mobility principles in which charged titanium dioxide (TiO_2, white) and carbon black pigment particles migrate through a non-polar dielectric fluid under an applied electric field. This evaluation synthesizes first-principles electro-optical physics with empirical register-level analysis across manufacturer datasheets (Waveshare 1.54inch e-Paper V2 Rev 2.0, Solomon Systech `SSD1681` Rev 0.13, Good Display GDEY0154D67 Rev 1.0), the Waveshare e-Paper GitHub repository, the `libdriver/ssd1681` driver specification, and seven Good Display firmware archives (2021–2025).

Summary of Core Engineering Findings:

- **On-Chip OTP Partial Refresh Support (`Display Mode 2`):** Programmed SSD1681 modules across both the 2019 (`SYX 1920` / GDEH0154D67) and 2020–2025 (GDEY0154D67) production generations contain pre-programmed 1-bit Partial Refresh waveforms in on-chip One-Time Programmable (OTP) memory assigned to **`Display Mode 2`**. Executing Command `0x22 → 0xFF` (or loading via `0xB9` and triggering via `0xCF`) invokes the temperature-compensated OTP partial waveform directly without requiring a 159-byte LUT transfer over SPI.
- **Production Exclusion of 2-Bit Grayscale Operation:** The 2-bit (4-level) grayscale routine distributed in late-revision vendor sample code operates by writing a synthetic temperature value of **+90.0°C (`0x5A, 0x00`)** to Register `0x1A` and issuing Command `0x22 → 0x91` to address an upper OTP waveform slot. This mechanism is unsuitable for production firmware because: (1) modules manufactured prior to late 2023 do not contain a 4-gray LUT at the +90.0°C OTP index; (2) overwriting Register `0x1A` bypasses real-

time thermal compensation across the rated 0°C to +50°C operating range; (3) intermediate gray states reside on the steep Kubelka-Munk optical transition slope; and (4) 2-bit pixel encoding occupies both 5,000-byte on-chip SRAM planes (`0x24` and `0x26`), precluding hardware differential partial refresh and requiring a 2.0-second full update on every write.

- **Production Approval of 1-Bit Bi-Tonal Full and Partial Refresh:** Standard 1-bit bi-tonal operation (full refresh via `0x22 → 0xF7` and 0.26-second non-flashing partial refresh via `0x22 → 0xFF`) meets all reliability and repeatability criteria when the firmware maintains a consistent data-entry mode (`0x11 → 0x03`), locks the border waveform (`0x3C → 0x80`) during partial updates, synchronizes the `OLD` (`0x26`) and `NEW` (`0x24`) RAM planes, and executes a full refresh every 5 to 10 partial updates to eliminate cumulative DC bias.

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2. Theoretical Background: Electrophoretic Particle Migration and Multi-Tonal Limitations

Electrophoretic displays are analog colloidal fluid suspensions. In a bi-tonal microcapsule, charged TiO_2 (white) and carbon black pigment particles are suspended in a non-polar hydrocarbon fluid. Unlike liquid crystal elements that modulate polarization in place, electrophoretic pigments translate physically across the microcapsule cell gap d.

2.1 Electrophoretic Mobility and Fluid Dynamics

When a potential difference V is applied across the source electrode and `VCOM`, particle drift velocity v is governed by electrophoretic mobility μ and electric field $E = V / d$:

$$\boldsymbol{v = \mu \cdot E = [\ (\zeta \cdot \varepsilon)\ /\ \eta(T)\ ] \cdot (\ V\ /\ d\ )}$$

- $\boldsymbol{\zeta}$ **(Zeta Potential):** Electrokinetic potential at the slipping plane of the suspended TiO_2 or carbon black particles.
- $\boldsymbol{\varepsilon}$ **(Dielectric Permittivity):** Permittivity of the suspending hydrocarbon dielectric fluid.
- $\boldsymbol{\eta(T)}$ **(Dynamic Viscosity):** Temperature-dependent shear viscosity of the suspending fluid.
- $\boldsymbol{V}$ **(Applied Potential Difference):** Net voltage between the SSD1681 source driver (`VSH1 = +15 V`, `VSH2 = +5 V`, `VSL = −15 V`, or `VSS = 0 V`) and `VCOM`.
- $\boldsymbol{d}$ **(Cell Gap Distance):** Microcapsule thickness between boundary electrodes (typically 15 to 30 µm).

In 1-bit bi-tonal operation, the time-integrated voltage impulse $\int V(t)\ \mathrm{d}t$ drives the pigment particles to physical saturation against the viewing boundary. Producing intermediate 2-bit grayscale states (Light Gray `01b` and Dark Gray `10b`) requires arresting particle transit mid-capsule prior to physical saturation.

2.2 Non-Linear Arrhenius Viscosity and Thermal Sensitivity

Fluid viscosity $\eta(T)$ follows an exponential Arrhenius relationship with absolute temperature T (in Kelvin), where E_a is the activation energy and R_g is the universal gas constant:

$$\boldsymbol{\eta(T) \propto \exp[\ E_a\ /\ (\ R_g \cdot T\ )\ ] \quad \Rightarrow \quad x(t,\ T) \propto [\ (\zeta \cdot \varepsilon)\ /\ d\ ] \cdot \exp[\ -E_a\ /\ (\ R_g \cdot T\ )\ ] \cdot \int V(t)\ \mathrm{d}t}$$

A waveform pulse calibrated to achieve 50% physical migration at **+22°C** over-drives particle displacement by approximately 34% at **+32°C** (shifting mid-tones toward saturated black) and under-drives displacement by approximately 32% at **+10°C** (yielding low-contrast light gray). Maintaining uniform

intermediate gray levels across the rated operating range (0°C to +50°C) requires temperature-indexed multi-level LUTs rather than a single fixed-temperature table.

2.3 Kubelka-Munk Non-Linear Optical Response

Particle displacement x does not map linearly to optical reflectance R. Optical absorption (K) and scattering (S) follow Kubelka-Munk radiative transfer theory, producing a sigmoidal reflectance response and non-linear perceptual lightness L^*:

$$K / S = (1 - R_\infty)^2 / (2 \cdot R_\infty) \quad \text{and} \quad L^* = 116 \cdot (R / R_0)^{1/3} - 16$$

Minor variations in effective pulse width (±2 ms) or ambient temperature place the pigment distribution within the steep transition region where $|\mathrm{d}R / \mathrm{d}t|$ is at a maximum, causing noticeable luminance variation across consecutive updates.

2.4 Hysteresis, Dual-Plane SRAM Allocation, and DC Charge Balance

Electrophoretic microcapsules exhibit strong state hysteresis: the optical reflectance produced by a short drive pulse depends on the preceding spatial distribution of the pigment particles (White → Dark Gray ≠ Black → Dark Gray). Furthermore, the SSD1681 contains exactly two 5,000-byte SRAM banks (`0x24` and `0x26`). Allocating both SRAM planes to represent a 2-bit grayscale value per pixel leaves no reference buffer for hardware differential partial updates. Unbalanced partial pulses can also accumulate residual DC charge across the dielectric fluid unless periodically neutralized by charge-balanced inversion waveforms.

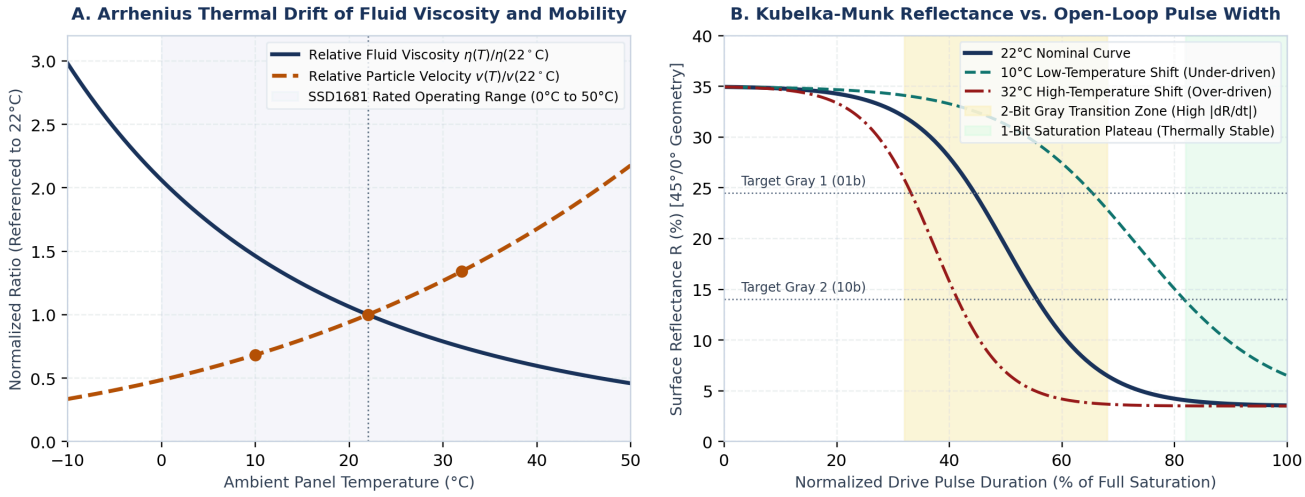


Figure 1: (A) Arrhenius thermal dependence of electrophoretic fluid viscosity $\eta(T)$ and particle drift velocity $v(T)$ normalized to 22°C. (B) Kubelka-Munk surface reflectance $R(\%)$ under 45°/0° geometry (referenced to the 35% White / 3.5% Dark / 10:1 Contrast Ratio specification in the Waveshare 1.54" V2 Datasheet, p. 25), comparing the 1-bit saturation plateaus with the 2-bit transition region.

Evaluation Criterion	2-Bit Analog Grayscale Behavior (4-Level)	1-Bit Bi-Tonal Behavior & Engineering Impact
Edge Definition	Susceptible to softened glyph boundaries; intermediate tones reduce local stroke contrast at 184 DPI.	Maintains full 10:1 contrast ratio ($R_{\mathrm{WS}} = 35\%$, $R_{\mathrm{DS}} = 3.5\%$) with sharp pixel-aligned strokes.
Thermal Stability	High sensitivity to ambient temperature when driven via fixed-temperature register overrides (`0x1A`).	Overdriven saturation plateaus maintain consistent black and white endpoints across 0°C to +50°C.

Evaluation Criterion	2-Bit Analog Grayscale Behavior (4-Level)	1-Bit Bi-Tonal Behavior & Engineering Impact
Ghosting & Hysteresis	High inter-frame hysteresis; requires full-screen clearing cycles on every frame update.	Hardware differential partial refresh (`0x24` NEW vs. `0x26` OLD) updates modified regions without flashing.
DC Bias & Lifespan	Increased risk of residual DC charge accumulation if asymmetric gray pulses are repeated.	Symmetric charge-balanced 1-bit waveforms preserve the rated 1,000,000-update / 5-year panel operating life.
Update Latency	Requires 1,500 to 2,000 ms per update with full-screen flashing.	Supports 260 to 300 ms non-flashing partial refresh (`Display Mode 2`) and 1,500 ms fast full refresh.
Controller Resource	Occupies both on-chip SRAM planes (`0x24` & `0x26`) and requires a 10,000-byte host buffer.	Requires a 5,000-byte host buffer and preserves Register `0x26` for hardware differential partial updates.

3. Empirical Revision History and Controller Register Equivalence

Waveshare has provided substantial value to the embedded engineering community by maintaining open-source, MIT-licensed C, Python, Arduino, and STM32 driver libraries (waveshareteam/e-Paper), English technical documentation, and level-shifted breakout modules. Because these software repositories support hardware revisions spanning nearly a decade, organizing the primary datasheets, panel markings, and code archives chronologically clarifies the architectural differences across the 1.54-inch 200×200 display family.

3.1 Chronological Hardware, Silicon, and Firmware Timeline (2017–2025)

Date / Code	Hardware / Document / Archive	Controller & LUT Type	Empirical Evidence & Technical Significance
2017-02-28	FPC Tooling Marking on `HINK-E0154A07-A1`	24-Pin 0.5 mm Pitch COG Footprint	Tooling date printed on the flexible printed circuit (FPC) of the target display, denoting the mechanical CAD layout date of the `HINK-E0154A07-A1` 24-pin flex circuit.
2018-07-17 to 2019-06-12	Waveshare `EPD_1in54.c` (V1 Driver, Rev V1.0 → V3.0) in e-Paper Repo	Solomon Systech `SSD1608` / Ilitek `IL3829` (**30-Byte LUT**)	Drives the first-generation 1.54" B/W panel (`HINK-E0154A05` / GDEP015OC1). Characterized in `EPD_1in54.c` by a **30-byte LUT** written to Register `0x32`, dummy-line register `0x3A`, gate-time register `0x3B`, and update sequence `0x22` → `0xC4`.
2019-04-04 to 2019-06-05	Solomon Systech `SSD1681.pdf` (Rev 0.10 → 0.13) & Waveshare V2 Datasheet	Solomon Systech `SSD1681` (**159-Byte WS / 36-Slot OTP**)	`SSD1681.pdf` (p. 2) logs initial IC release Rev 0.10 on `09-May-19` and Rev 0.13 on `05-Jun-19`. Waveshare's V2 datasheet (p. 3) logs Rev 1.0 on `Apr.04.2019` and Rev 2.0 on `Jun.05.2019`.

Date / Code	Hardware / Document / Archive	Controller & LUT Type	Empirical Evidence & Technical Significance
2019 Week 20 (SYX 1920)	Target Panel Lot Marking (`HINK-E0154A07-A1`)	SSD1681 (Early 2019 OTP Configuration)	Lot date code `1920` (Year 2019, Week 20, mid-May 2019). Corresponds to the initial production period of the SSD1681 controller and 1.54" V2 hardware (GDEH0154D67 class), containing 1-bit B/W Full (Mode 1) and Partial (Mode 2) OTP waveforms.
2019-06-11	Waveshare `EPD_1in54_V2.c` (V1.0) in e-Paper Repo	SSD1681 (**159-Byte External SPI LUT**)	Waveshare releases `EPD_1in54_V2.c` with explicit 159-byte arrays `WF_Full_1IN54[159]` and `WF_PARTIAL_1IN54_0[159]` written over SPI (`0x32`, `0x3F`, `0x03`, `0x04`, `0x2C`) to provide uniform waveform timing independent of module OTP contents.
2020-11-23	Good Display GDEY0154D67.pdf (Rev 1.0 Specification)	SSD1681 (Adds 1.5 s Fast Update Rating)	Good Display publishes the `GDEY0154D67` specification, rating Full (`2 s`), Fast (`1.5 s`), and Partial (`0.26 s`) refresh times (p. 22), while specifying the optical gray rating as `2Grey Level` (1-bit B/W, p. 29).
2021-05-18	Good Display `RP-GDEY0154D67-P-210518.rar` (RPi Pico)	SSD1681 (OTP Mode 1 `0xF7` & Mode 2 `0xFF`)	Archive suffix `-P` (Full + Partial). Uses internal temperature sensing (`0x18 → 0x80`) and OTP Mode 2 partial refresh (`0x22 → 0xFF`). Does not include Fast (`0x64`) or 4-Gray (`0x5A`) register overrides.
2022-10-11 to 2022-10-13	Good Display `S-GDEY0154D67-210414.rar` (`-FP-20221011`) & `A32/A8266/AU-FP-20221013`	SSD1681 (Adds `0x1A → 0x64` +100°C Fast Slot)	Archive suffix `-FP` (Full + Fast + Partial). Introduces `EPD_HW_Init_Fast()` using synthetic temperature `0x64` (+100.0°C) and `0x22 → 0x91`. No 4-Gray routine is present in the 2022 archives.
2025-02-12	Good Display `S-GDEY0154D67.rar` & `A32-GDEY0154D67.rar` (`-FP4G-20250212`)	SSD1681 (Adds `0x1A → 0x5A` +90°C 4-Gray Slot)	Archive suffix `-FP4G` (Full + Fast + Partial + 4-Gray). Introduces `EPD_HW_Init_4G()` using synthetic temperature `0x5A` (+90.0°C) and `0x22 → 0x91`, noting in `main.c`: `/* Onder [Older] versions do not support this feature */`.

3.2 Controller Register Equivalence and OTP Provenance Boundaries

1. **Silicon Register Equivalence:** Direct comparison of Solomon Systech's `SSD1681.pdf` (Rev 0.13, Section 7, pp. 20–33), Waveshare's `1.54_inch_epd_datasheet_v2.pdf` (Section 5, pp. 20–24), and Good Display's `GDEY0154D67.pdf` (Section 6, pp. 10–21) confirms identical command opcodes, bit definitions, and timing parameters. The mechanical drawing title block on Page 7 of `GDEY0154D67.pdf` explicitly specifies `LCD controller/driver: SSD1681`.
2. **Physical Form Factor and Reference Circuitry:** Both `1.54_inch_epd_datasheet_v2.pdf` and `GDEY0154D67.pdf` define the same 24-pin 0.5 mm-pitch FPC pinout, 37.32×31.80 mm glass outline, 1.18 mm thickness, external boost MOSFET topology (`Si1304BDL / Si1308EDL` with `MBR0530` Schottky diodes), and `Eye-One Pro Spectrophotometer` optical test standard.
3. **Boundary of Empirical Evidence on OTP Programming:** While SPI diagnostic registers (`0x2D`, `0x2E`, and `0x34`/`0x35`) allow firmware to verify the exact bytes and CRC checksum present in a module's OTP

memory, the silicon registers do not record which entity programmed the OTP tables or the date and production stage at which programming occurred. Consequently, firmware must rely on runtime register verification rather than external lot labels to determine OTP capability.

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4. SSD1681 OTP Architecture, Waveform Configuration Map, and Hexadecimal Decode Matrix

Interrogating an SSD1681 module over 4-wire SPI returns raw hexadecimal bytes that must be decoded against the silicon register specification to determine the active waveform configuration and OTP structure.

4.1 SSD1681 OTP Waveform Slot Allocation and Display Mode Architecture

The SSD1681's 5,832-byte OTP memory accommodates up to **36 Waveform Settings (`WS0–WS35`, 159 bytes each)** paired with **36 Temperature Ranges (`TR0–TR35`, 3 bytes each)** (`SSD1681.pdf` Section 6.10, p. 18). The controller distinguishes between full-refresh and partial-refresh OTP waveform slots using the Display Mode selection bitmap in **OTP Register `0x37` (read back via Command `0x2D`, Bytes C through G)**:

- **`Display Mode 1` (Bit = `0` in `0x37` / `0x2D`):** Designates standard Full Refresh waveform slots (and, on later OTP revisions, synthetic high-temperature slots at +90.0°C and +100.0°C used for 4-Gray and Fast Refresh). Mode 1 maps all four `(0x26, 0x24)` SRAM bit combinations directly to `LUT0..LUT3`.
- **`Display Mode 2` (Bit = `1` in `0x37` / `0x2D`):** Designates **Hardware Partial Refresh** waveform slots. Programmed SSD1681 panels (`GDEH0154D67`, `GDEY0154D67`, and `HINK-E0154A07-A1`) include dedicated partial-refresh temperature bands in OTP with their corresponding Display Mode bit set to `1`. Issuing Command `0x22 → 0xFF` (or `0xB9` followed by `0xCF`) directs the state machine to select the OTP Mode 2 partial-refresh waveform matching the current ambient temperature.

Operating Mode	Temp Register `0x1A`	LUT Load Cmd (`0x22`)	Fire Cmd (`0x22`)	Hardware State Machine Execution (`SSD1681.pdf` p. 26)
1-Bit Full Refresh (Auto OTP)	Auto-sensed via `0x18 → 0x80` (−25°C to +50°C)	*Included in `0xF7`* (or pre-load via **`0xB1`**)	**`0xF7`** + `0x20` (~2.0 s)	Reads internal temperature sensor → searches OTP `TR0..TR35` where **Mode = 0 (Mode 1)** → loads 1-bit Full LUT → drives panel. **Supported across all programmed revisions (2019–2025).**
1-Bit Partial Refresh (Auto OTP)	Auto-sensed via `0x18 → 0x80` (0°C to +50°C)	*Included in `0xFF`* (or pre-load via **`0xB9`**)	**`0xFF`** + `0x20` (~0.26 s)	Reads internal temperature sensor → searches OTP `TR0..TR35` where **Mode = 1 (Mode 2)** → loads OTP Partial LUT → drives differential update (`0x26` OLD vs. `0x24` NEW). **Supported across all programmed revisions.**
1-Bit Fast Refresh (Synthetic +100°C OTP)	Manual override: `0x1A → 0x64, 0x00` (**+100.0°C**)	**`0x91`** + `0x20` (Bypasses sensor read)	**`0xC7`** + `0x20` (~1.5 s)	Searches OTP `TR0..TR35` (Mode 1) for **+100°C** → loads single-flash 1.5 s Fast LUT → triggers with `0xC7` without re-reading the sensor. **Supported on 2022+ `GDEY0154D67` (`-FP` & `-FP4G`).**

Operating Mode	Temp Register `0x1A`	LUT Load Cmd (`0x22`)	Fire Cmd (`0x22`)	Hardware State Machine Execution (`SSD1681.pdf` p. 26)
2-Bit (4-Gray) Refresh (Synthetic +90°C OTP)	Manual override: `0x1A → 0x5A, 0x00` (**+90.0°C**)	**`0x91`** + `0x20` (Bypasses sensor read)	**`0xC7`** + `0x20` (~2.0 s)	Searches OTP `TR0..TR35` (Mode 1) for **+90°C** → loads 4-Gray LUT → drives 4 levels from `(0x26, 0x24)` planes. **Supported only on late 2023–2025 `GDEY0154D67 (-FP4G)`.**
External SPI Custom LUT (Full / Gray)	Ignored (LUT written directly to `0x32`)	Write 153 B to **`0x32`** (+ `0x3F, 0x03, 0x04, 0x2C`)	**`0xC7`** + `0x20`	Bypasses OTP lookup and executes the 159-byte waveform written by the host MCU to Register `0x32` in Display Mode 1. **Supported on all SSD1681 panels (including 2019 `SYX 1920`).**
External SPI Custom LUT (Partial)	Ignored (LUT written directly to `0x32`)	Write 153 B to **`0x32`** + `0x37` (`Byte F = 0x40`)	**`0xCF`** + `0x20`	Used by Waveshare's `EPD_1in54_V2.c` (`WF_PARTIAL_1IN54_0`). Enables RAM Ping-Pong via Register `0x37` and executes the custom partial LUT in Display Mode 2.

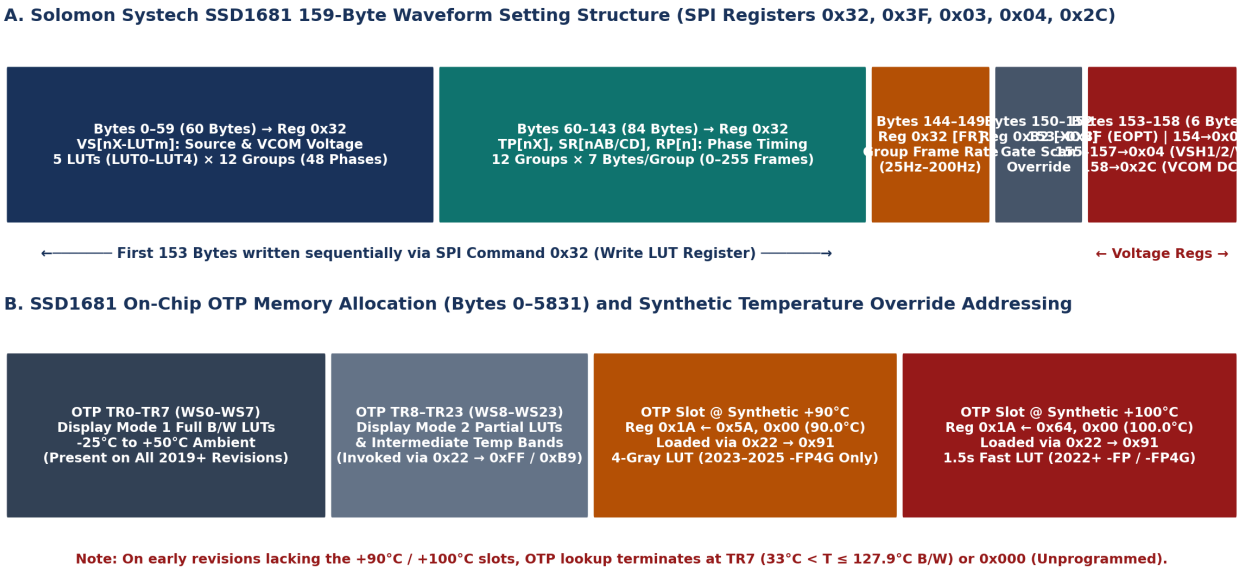


Figure 2: (A) Byte-level mapping of a 159-byte Waveform Setting across SSD1681 SPI registers `0x32` (153 bytes), `0x3F` (1 byte), `0x03` (1 byte), `0x04` (3 bytes), and `0x2C` (1 byte). (B) On-chip OTP layout showing Mode 1 (Full), Mode 2 (Partial), and the +90.0°C / +100.0°C synthetic-temperature slots.

4.2 Hard Reference Hexadecimal Decode Matrix (`Chip_ID`, `Option Matrix`, `User_ID`, `CRC`)

As specified in the `libdriver/ssd1681` reference (`ssd1681-otp-fingerprinting.md`) and `SSD1681.pdf` (pp. 21, 24, 28–30), the 153-byte LUT register `0x32` is write-only (`R/W# = 0`), meaning raw waveform bytes cannot be read directly out of OTP over SPI. Instead, firmware identifies a module's silicon controller, VCOM calibration, waveform mode allocation, and OTP revision by reading five verification registers (`0x2F`, `0x0A`, `0x2D`, `0x2E`, and `0x34`/`0x35`) and decoding the returned hexadecimal bytes against the reference matrix below:

SPI Read Cmd & Byte	Register / Field Name	Unprogrammed / POR Hex	Known Valid Hexadecimal Signatures	Hexadecimal Decode Specification
0x2F Byte 0	**Status Bit Read / Chip_ID** (SSD1681.pdf p. 29)	0x01 (Binary 0000_0001b)	**0x01** (Idle, Normal VCI, HV Ready, **Chip ID = 01b**) **0x05** (Busy = 1, Chip ID = 01b)	Bits [1:0] = **01b (0x01)** confirms Solomon Systech **SSD1681** silicon. Bit [2] = Busy flag (0 = Ready, 1 = Busy). Bit [4] = VCI low (<2.3 V if 1). Bit [5] = High-voltage supply not ready (if 1).
0x0A Bytes 0–3	**Initial Code Setting Read** (libdriver 0x0A, p. 21)	00 00 00 00	Typically 00 00 00 00 or programmed gate/source trim bytes	Verifies the 4-byte Initial Code OTP block loaded via Command 0x08 / 0x09.
0x2D Byte 0 (A)	**VCOM OTP Selection** (0x37 Byte A, p. 28 & 30)	0x00	**0x00** = Default OTP VCOM **0x80** = Spare OTP VCOM	Bit A[7] = 0 selects the primary OTP VCOM byte; Bit A[7] = 1 selects the spare VCOM slot.
0x2D Byte 1 (B)	**Programmed VCOM Value** (0x2C mirror, p. 28)	0x00 (Unprogrammed)	Waveshare Full LUT: **0x20** (−0.8 V) Waveshare Part LUT: **0x28** (−1.0 V) Typical OTP Range: **0x3C–0x58** (−1.5 V to −2.2 V)	Convert hex to decimal and evaluate: $V_{\mathrm{COM}} = -(\mathrm{Byte\ B} / 4) \times 0.1\ \mathrm{V}$. Example: 0x28 (40) → **−1.0 V**; 0x50 (80) → **−2.0 V**. Copy this byte into lut[158] (0x2C) when constructing custom SPI LUTs.
0x2D Bytes 2–6 (C..G)	**Display Mode Option Matrix (WS0..WS35)** (0x37 Bytes B–F, p. 30)	00 00 00 00 00 (Waveshare Partial override: 00 00 00 00 40)	**Typical Mode 1 + Mode 2 Layout:** Byte C = 0x00 (WS0..7 = Mode 1 Full), Byte D = 0xFF (WS8..15 = Mode 2 Partial), Byte G[6] = 1 (0x40, Ping-Pong ON)	36-bit bitmap mapping each OTP waveform slot WS[0..35]: • Bit = **0 ⇒ Display Mode 1** (Full / Fast / 4-Gray). • Bit = **1 ⇒ Display Mode 2** (OTP Partial Refresh). • **Byte G (G) Bit 6 (0x40):** 1 = Hardware RAM Ping-Pong enabled for Mode 2 Partial Refresh.
0x2D Bytes 7–10 (H..K)	**4-Byte Waveform Version ID** (0x37 Bytes G–J, p. 28 & 30)	00 00 00 00	• **Gen 1 (2019 GDEH0154D67 / SYX 1920):** Base Mode 1 + Mode 2 Partial (lacks 0x5A/0x64). • **Gen 2 (2020–22 GDEY0154D67 -FP):** Adds 0x64 (+100°C Fast). • **Gen 3 (2023–25 GDEY0154D67 -FP4G):** Adds 0x5A (+90°C 4-Gray).	Stores the 4-byte waveform version identifier programmed into Register 0x37. Matching Bytes 7–10 across two modules indicates identical OTP waveform table revisions.

SPI Read Cmd & Byte	Register / Field Name	Unprogrammed / POR Hex	Known Valid Hexadecimal Signatures	Hexadecimal Decode Specification
0x2E Bytes 0–9 (A..J)	**10-Byte User_ID Register** (0x38, p. 28 & 30)	00 00 00 00 00 00 00 00 00 00 (or FF..FF)	00..00 (unprogrammed) or 10-byte batch identifier	Read via Command 0x2E (ssd1681_get_user_id). Returns the 10-byte identifier written to OTP Register 0x38, if populated.
0x34 → 0x35 Bytes 0–1	**16-Bit Hardware CRC Checksum** (libdriver 0x35, p. 29)	Varies by RAM/OTP state	Deterministic 16-bit big-endian checksum A[15:0]	**Verification of Upper OTP Slots:** Load the 25°C LUT (0x1A → 0x19, 0x00 + 0x22 → 0x91), execute 0x34/0x35, and record the 16-bit CRC. Then load +90°C (0x1A → 0x5A, 0x00 + 0x22 → 0x91) and measure update duration and CRC. If the +90°C slot is unprogrammed, BUSY either deasserts in <50 ms or falls back to TR7 (33°C to 127.9°C B/W LUT).

5. Production Readiness Assessment and Optimal Register Configurations

Deploying reliable embedded display firmware requires distinguishing between undocumented register overrides and specification-compliant production configurations.

5.1 Technical Comparison of Experimental 2-Bit Grayscale vs. Production 1-Bit Partial Refresh

- **2-Bit Grayscale Assessment — Experimental / Laboratory Use Only:** Whether driven via the 0x1A → 0x5A (+90.0°C) OTP override or an external 159-byte SPI LUT (0x32), 2-bit grayscale bypasses ambient temperature compensation, exhibits visible luminance drift when ambient temperature varies by ±5°C, requires 10,000 bytes of SRAM (both 0x24 and 0x26), and performs a 2.0-second flashing refresh on every write. For small-screen typography at 184 DPI, properly hinted 1-bit bitmap fonts (with 2-pixel stem widths at ≥14 pt) provide higher contrast and legibility.
- **1-Bit Partial Refresh Assessment — Production-Ready:** The SSD1681 hardware partial refresh (Display Mode 2, 0.26-second non-flashing update) is fully suitable for production deployment when firmware enforces two register-management rules:
  1. **Maintain Consistent Data Entry Mode (0x11 → 0x03) Across Resets:** In several vendor sample scripts, EPD_Dis_Part() pulses hardware RES# (which resets Register 0x11 to its power-on default of 0x03) without restoring the prior 0x11 → 0x01 setting, inverting the Y-axis coordinate mapping. Standardizing on **0x11 → 0x03 (X increment, Y increment)** across both initialization and partial updates places memory address 0 consistently at top-left origin (0,0).
  2. **Synchronize Dual SRAM Planes (0x24 NEW and 0x26 OLD) and Enforce Periodic Full Refresh:** Writing the updated framebuffer to 0x24 during partial refresh and synchronizing 0x26 ensures accurate bit-by-bit differential comparisons even on modules where OTP Ping-Pong is not enabled. As specified on Page 23 of GDEY0154D67.pdf: *"During the fast refresh or partial refresh of the electronic paper, it is recommended to add a full-screen refresh after 5 consecutive operations to reduce the accumulation of afterimages on the screen."*

5.2 Optimal Production Register Configurations for SSD1681 Modules

Operational Profile	SPI Register Sequence	Refresh Time & Power	Production Usage Policy & Notes
Profile 1: Standard 1-Bit Full Refresh (Baseline & Anti-Ghosting)	`0x12` (SWRESET) `0x01 → C7 00 00` (200 gates) `0x11 → 03` (X+, Y+ top-left origin) `0x44 → 00 18` | `0x45 → 00 00 C7 00` `0x3C → 05` (Border follows LUT1) `0x18 → 80` (Internal temp sensor) Write 5,000 B to `0x24` **and** `0x26` **`0x22 → F7`**, `0x20`	**2.0 sec** ~4.5 mW typ. (<0.003 mW sleep)	**Primary Baseline Specification:** Uses the internal temperature sensor across 0°C to +50°C. Writing the initial image to both `0x24` and `0x26` initializes the reference buffer for subsequent partial updates.
Profile 2: Non-Flashing 1-Bit Partial Refresh (UI & Text Updates)	Hardware Reset (RES# 10 ms) *or* wake **`0x11 → 03`** (Maintain X+, Y+ addressing) **`0x3C → 80`** (Lock border VBD = VCOM) Set window `0x44`, `0x45` & cursor `0x4E`, `0x4F` Write new data to `0x24` **`0x22 → FF`**, `0x20` (Copy new data to `0x26` for next frame)	**0.26–0.30 sec** Non-flashing	**Dynamic UI Profile:** Suitable for updating text, counters, or status indicators. Maintain an update counter that executes **Profile 1 (Full Refresh)** after every **5 to 10 partial updates** to prevent ghost accumulation and preserve DC balance.
Profile 3: 1.5 s Single-Flash Fast Full Refresh	`0x12` (SWRESET) `0x18 → 80` | `0x22 → B1`, `0x20` **`0x1A → 64 00`** (+100°C Fast OTP) **`0x22 → 91`**, `0x20` Write 5,000 B to `0x24` **`0x22 → C7`**, `0x20`	**1.5 sec** Single flash	Supported on 2022+ `GDEY0154D67` modules. Because this mode overrides temperature register `0x1A`, restrict operation to ambient room temperatures (15°C to 35°C) and execute Profile 1 every 5 cycles.
Profile 4: Mandatory Deep Sleep & Storage State	**`0x10 → 01`** (Deep Sleep Mode 1: retains RAM, 1 µA) *or* **`0x10 → 03`** (Deep Sleep Mode 2: drops RAM, 0.7 µA)	**0 ms** 0.7–1.0 µA	**Hardware Protection Rule:** Transmit `0x10 → 0x01` immediately after `BUSY` transitions LOW to disable the high-voltage (+15 V / −15 V / +20 V) charge pumps. Clear the panel to white prior to extended storage.

6. Cross-Revision Documentation Notes and DPI Geometry Analysis

Because the available datasheets and driver repositories support multiple controller generations and evaluation boards, several minor documentation differences appear across revisions. Resolving these details mathematically and at the register level simplifies firmware integration:

6.1 Geometric Derivation of the 183.66, 184, and 188 DPI Specifications

- **183.66 and 184.06 DPI (27.60 mm Active Area Geometry):** A square array of 200×200 pixels has a diagonal of $\sqrt{(200^2 + 200^2)} = 282.8427$ pixels. Dividing by the nominal 1.54 inch diagonal yields **183.664 DPI**. Similarly, the $27.60\ \text{mm} \times 27.60\ \text{mm}$ active area (0.138 mm pixel pitch) specified on

the Waveshare 1.54inch e-Paper product page and in `GDEY0154D67.pdf` (p. 6) evaluates to $25.4\ \text{mm/inch} / 0.138\ \text{mm} = \mathbf{184.06\ DPI}$ (rounded to `Dpi: 184`).

- **188.15 DPI (27.00 mm Active Area Geometry):** In Waveshare's V2 Datasheet (p. 4), the active area is tabulated as $27.00\ \text{mm} \times 27.00\ \text{mm}$ with a $0.135\ \text{mm}$ pixel pitch, which evaluates to $25.4\ \text{mm/inch} / 0.135\ \text{mm} = \mathbf{188.15\ DPI}$ (rounded to `Dpi: 188`). Both figures describe the same 200×200 pixel array, differing only by whether the active perimeter tolerance is measured at $27.00\ \text{mm}$ (1.503 inch diagonal) or $27.60\ \text{mm}$ (1.537 inch diagonal).

6.2 Cross-Revision Notes Across Datasheets and Sample Code

Source Reference	Section / File	Technical Context & Firmware Integration Guidance
Waveshare Repository	`EPD_1in54.c` Lines 28–56 & `Debug.h`	**Retired Controller Macros & Debug Wrapper:** The commented-out `#define` macros (`PANEL_SETTING 0x00`, `PLL_CONTROL 0x30`, etc.) in the V3.0 header of `EPD_1in54.c` were retained from an earlier UltraChip `UC8151` driver template during the `2019-06-12` repository refactoring. The `Debug(...)` macro in `Config/Debug.h` is an optional UART `printf` wrapper.
Waveshare V2 Datasheet	Page 1 (Cover) & Page 19 (Flowchart)	**Template Date & Sequence Parameter:** Page 1 retains the V1 document template date (`2017/06/05`), whereas Page 3 records the V2 revision history (`Rev 1.0: Apr.04.2019`, `Rev 2.0: Jun.05.2019`). On Page 19, when overriding temperature register `0x1A` externally, firmware must issue `0x22 → 0x91` (load LUT without re-reading the internal sensor) rather than `0xB1`.
Good Display `GDEY0154D67.pdf`	Page 9, Note 5-4 (BUSY Pin)	**`BUSY` Pin Polarity:** Note 5-4 describes `BUSY` as active-low (a convention used on UltraChip controllers). On all Solomon Systech SSD1681 modules, `BUSY` is **Active HIGH (`1` = Busy, `0` = Ready)**, consistent with `SSD1681.pdf` (p. 8) and the C driver implementations.
Good Display Partial Demo	`Display_EPD_W21.c` (`EPD_Dis_Part`)	**Restoring Register `0x11` After Hardware Reset:** Because pulsing hardware `RES#` restores Register `0x11` to its power-on default (`0x03` = X+, Y+), firmware should explicitly configure Register `0x11` after any reset pulse so full and partial updates share identical coordinate axes.

7. Laboratory Methodology for Measuring Reflectance and Profiling Particle Migration

To empirically characterize optical reflectance $R(V,\ t,\ T)$ across the operating temperature range (−10°C to +50°C) and map particle migration per LUT configuration, the following test apparatus extends the optical measurement standard defined in Section 6 of Waveshare's V2 Datasheet (pp. 25–26):

- **45°/0° Bidirectional Spectrophotometry:** Illuminate the display at **45° incidence** using a regulated D65 (6500 K, 1,000 Lux) ring source, and measure diffuse reflectance normal to the glass surface (**0° detection angle**, ≤2° acceptance cone) using an **X-Rite Eye-One Pro (i1Pro) Spectrophotometer** calibrated against a 99% $BaSO_4$ / Spectralon diffuse white reference standard:

$$\boldsymbol{R = R_{\text{reference}} \cdot \left(L_{\text{panel}} / L_{\text{reference}} \right)} \quad \textbf{and} \quad \boldsymbol{CR = R_{\text{white}} / R_{\text{dark}}}$$

- **Time-Resolved High-Speed Telecentric Imaging (240 fps):** Couple a global-shutter monochrome machine-vision camera (240 fps, 12-bit linear raw output) with a telecentric macro lens to record pixel reflectance $R(t)$ across each 5 ms to 20 ms frame interval while the SSD1681 executes a test LUT from Register `0x32`.
- **Peltier Thermal Sweep (−10°C to +50°C) and 16-Transition Matrix:** Mount the panel on a closed-loop Peltier thermal stage inside a dry-nitrogen enclosure, stepping from −10°C to +50°C in 5°C increments (verified against internal sensor register `0x1B`) while sweeping Phase Length `TP[nX]` (1 to 64 frames) and Source Voltages `0x04` (`VSH1`, `VSH2`, `VSL`) across all 16 state transitions $(G_{\text{initial}} \rightarrow G_{\text{target}}) \in \{0, 1, 2, 3\} \times \{0, 1, 2, 3\}$.

---

8. Waveshare 1.54-Inch (200×200 Pixel) Display Lineup and Alternative Reflective Technologies

The table below catalogs **exclusively the 1.54-inch 200×200 pixel e-Paper variants** offered by Waveshare, detailing each model's physical form factor (raw glass panel, PCB module, wireless enclosure, or MCU development board), hardware version (`V1` vs. `V2`), color palette, controller IC, and refresh ratings. *(Waveshare's 1.54inch e-Paper (C) [152×152 Yellow/Black/White] and 1.54inch e-Paper (E) [240×240 Spectra 6] are omitted because their native resolutions are not 200×200.)*

8.1 Complete Matrix of Waveshare 1.54-Inch (200×200 Pixel) E-Paper Variants

Waveshare 1.54" (200×200) Model	Display Type & Form Factor	Version & Colors	Controller IC & Panel Marking	Refresh Time (Full / Fast / Part)	LUT & Integration Notes
1.54inch e-Paper (Raw) & Module (V1)	Raw 24-pin FPC Glass *or* 3.3V/5V PCB Module	**V1** Black, White (2-Color)	`SSD1608` / `IL3829` (`HINK-E0154A05` / `GDEP015OC1`)	2.0 s Full — 0.3 s Part	Driven by `EPD_1in54.c` (30-byte LUT to `0x32`, `0x22` → `0xC4`). 1-bit B/W only; superseded by V2.
1.54inch e-Paper V2 (Raw) & Module V2	Raw 24-pin FPC Glass *or* 3.3V/5V PCB Module (PH2.0 8-pin)	**V2** Black, White (2-Color)	`SSD1681` (`HINK-E0154A07-A1` / GDEH0154D67 / GDEY0154D67)	2.0 s Full 1.5 s Fast 0.26 s Part	Driven by `EPD_1in54_V2.c` (159-byte LUT). Provides 0.26 s hardware partial refresh.
1.54inch e-Paper (B) (Raw) & Module (B) V1	Raw 24-pin FPC Glass *or* 3.3V/5V PCB Module	**(B) V1 Red, Black, White** (3-Color)	`IL0376F` / `UC8151` (`GDEW0154Z04` / `WFT0000CZ04`)	8.0 s Full — No Partial	Driven by `EPD_1in54b.c` (commands `0x10`/`0x13`/`0x12`). Superseded by (B) V2.
1.54inch e-Paper (B) V2 & Module (B) V2	Raw 24-pin FPC Glass *or* 3.3V/5V PCB Module	**(B) V2 Red, Black, White** (3-Color)	`SSD1681` (`GDEH0154Z90` / `GDEY0154Z90`)	8.0–14.0 s Full — No Partial	Driven by `EPD_1in54b_V2.c`. Uses the **SSD1681** controller with `0x24` assigned to B/W RAM and `0x26` assigned to Red RAM (`~redimage`).

Waveshare 1.54" (200×200) Model	Display Type & Form Factor	Version & Colors	Controller IC & Panel Marking	Refresh Time (Full / Fast / Part)	LUT & Integration Notes
1.54inch e-Paper (G) (Raw & Module)	Raw 24-pin FPC Glass *or* 3.3V/5V PCB Module	**(G) V1 & V2** **Red, Yellow, Black, White** (4-Color)	`JD79661` (`GDEY0154F51`)	20.0 s Full 15.0 s Fast No Partial	Driven by `EPD_1in54g.c` / `EPD_1in54g_V2.c`. Encodes 2 bits/pixel (`00`=Black, `01`=White, `10`=Yellow, `11`=Red) in a 10,000-byte buffer.
1.54inch NFC-Powered (BW / BB / G)	Battery-Free ABS Enclosure (Energy-Harvesting NFC)	**BW** (2-Color) **BB** (R/B/W) **G** (R/Y/B/W)	`SSD1681` (BW/BB) or `JD79661` (G) + NFC Harvester	2 s (BW) 8 s (BB) 15 s (G)	Harvests RF power and transfers 200×200 frames wirelessly from an NFC-enabled smartphone or ST25R3911B reader.
ESP32-S3-ePaper-1.54 (V1 & V2) & RP2350-ePaper-1.54	All-in-One MCU Dev Board (Onboard 200×200 EPD)	**V1 / V2** B/W (SSD1681) or 4-Color (G)	`SSD1681` (B/W) or `JD79661` (1.54G) + ESP32-S3 / RP2350A	2.0 s Full 1.5 s Fast 0.26 s Part	ESP32-S3 board integrates an onboard **SHTC3 temperature/humidity sensor**, PCF85063 RTC, audio codec, and LiPo charger.

8.2 Alternative Display Technologies for Fast Grayscale or Multi-Color Rendering

As noted in Waveshare's ESP32 E-Paper & RLCD Tutorial (*"Choose E-Paper when the image is largely static and standby time is the priority. Choose RLCD when the content must update in real time"*), applications requiring anti-aliased multi-bit typography or color in a 1.54-inch 200×200 footprint without multi-second refresh cycles can evaluate two reflective alternatives:

- **Reflective Memory-in-Pixel (MIP) LCDs (1.54" 200×200 & 1.28" 176×176):** Displays such as the HannStar HSD015GPN3 (1.54" 200×200 Reflective MIP LCD), JDI LPM013M126A (1.28" 176×176 8-/64-Color MIP LCD), and Sharp LS013B7DH03 operate without a backlight, consume 2 to 15 µW in static hold, and refresh in **10 to 20 ms (30 to 60 Hz)** without ghosting across −20°C to +70°C.
- **Sitronix `ST7302` Reflective 4-Gray LCDs & Transflective IPS TFTs:** Provide hardware-stable 2-bit (4-level) grayscale or 16-bit RGB565 color suitable for sub-pixel font anti-aliasing at 30 to 60 Hz.

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9. Reference C Implementation: OTP Hexadecimal Decoder, Production 1-Bit Driver, and 2-Bit Grayscale Override

The self-contained C driver below provides: (1) the bidirectional 4-wire SPI read interface and OTP register decoder (`SSD1681_RunForensicDump`); (2) the experimental **2-Bit Grayscale Synthetic Temperature Override** (`SSD1681_Init_4Gray_OTP_Override` and `SSD1681_Display_4Gray_OTP` using `0x1A → 0x5A, 0x00` and bitwise-inverted dual-plane splitting); and (3) the **production 1-Bit Full, Fast, and Partial Refresh driver** with automatic 5-cycle anti-ghosting refresh.

```
#include <stdint.h>
#include <stdbool.h>
#include <stdio.h>
```

```c
/* Hardware GPIO & Timer Hooks */
extern void EPD_CS_Set(uint8_t level);
extern void EPD_DC_Set(uint8_t level);
extern void EPD_RST_Set(uint8_t level);
extern void EPD_SCL_Set(uint8_t level);
extern void EPD_SDA_Set(uint8_t level);
extern void EPD_SDA_ModeInput(void);
extern void EPD_SDA_ModeOutput(void);
extern uint8_t EPD_SDA_Read(void);
extern uint8_t EPD_BUSY_Read(void);
extern void EPD_DelayMs(uint32_t ms);

/* ============================================================================
 * PART 1: Low-Level 4-Wire SPI Write & Bidirectional Read (SSD1681.pdf p. 11)
 * ============================================================================ */
static void SSD1681_SpiWriteByte(uint8_t val) {
    EPD_SDA_ModeOutput();
    for (uint8_t i = 0; i < 8; i++) {
        EPD_SCL_Set(0);
        EPD_SDA_Set((val & 0x80) ? 1 : 0);
        EPD_SCL_Set(1);
        val <<= 1;
    }
    EPD_SCL_Set(0);
}

static uint8_t SSD1681_SpiReadByte(void) {
    uint8_t val = 0;
    EPD_SDA_ModeInput();
    for (uint8_t i = 0; i < 8; i++) {
        EPD_SCL_Set(1);
        EPD_SCL_Set(0); /* Data shifted out on falling edge of SCL (<= 2.5 MHz) */
        val = (uint8_t)((val << 1) | (EPD_SDA_Read() & 0x01));
    }
    return val;
}

void SSD1681_WriteCmd(uint8_t cmd) {
    EPD_CS_Set(0); EPD_DC_Set(0);
    SSD1681_SpiWriteByte(cmd);
    EPD_CS_Set(1);
}

void SSD1681_WriteData(uint8_t data) {
    EPD_CS_Set(0); EPD_DC_Set(1);
    SSD1681_SpiWriteByte(data);
    EPD_CS_Set(1);
}

void SSD1681_ReadReg(uint8_t cmd, uint8_t *buf, uint16_t len) {
    EPD_CS_Set(0); EPD_DC_Set(0);
    SSD1681_SpiWriteByte(cmd);
    EPD_DC_Set(1);
    for (uint16_t i = 0; i < len; i++) buf[i] = SSD1681_SpiReadByte();
    EPD_SDA_ModeOutput();
    EPD_CS_Set(1);
}

void SSD1681_WaitBusy(void) {
    while (EPD_BUSY_Read() == 1) EPD_DelayMs(2); /* 1 = Busy, 0 = Ready */
}

/* ============================================================================
```

```
 * PART 2: OTP Hex Dump & Register Decoder (0x2F, 0x0A, 0x2D, 0x2E, 0x35)
 * ========================================================================== */
void SSD1681_RunForensicDump(void) {
    uint8_t status = 0, init_code[4] = {0}, opt[11] = {0}, uid[10] = {0}, tmp[2] = {0}, crc[2] =
{0};

    EPD_RST_Set(0); EPD_DelayMs(10); EPD_RST_Set(1); EPD_DelayMs(10);
    SSD1681_WaitBusy();
    SSD1681_WriteCmd(0x12); SSD1681_WaitBusy();

    /* Sense temperature & load default Mode 1 LUT */
    SSD1681_WriteCmd(0x18); SSD1681_WriteData(0x80);
    SSD1681_WriteCmd(0x22); SSD1681_WriteData(0xB1);
    SSD1681_WriteCmd(0x20); SSD1681_WaitBusy();

    /* Read all 5 diagnostic registers + compute hardware CRC */
    SSD1681_ReadReg(0x2F, &status, 1);
    SSD1681_ReadReg(0x0A, init_code, 4);
    SSD1681_ReadReg(0x2D, opt, 11);
    SSD1681_ReadReg(0x2E, uid, 10);
    SSD1681_ReadReg(0x1B, tmp, 2);
    SSD1681_WriteCmd(0x34); SSD1681_WaitBusy(); /* Compute CRC */
    SSD1681_ReadReg(0x35, crc, 2);

    int16_t raw12 = (int16_t)(((uint16_t)tmp[0] << 4) | (tmp[1] >> 4));
    if (raw12 & 0x800) raw12 |= 0xF000;
    float temp_c = (float)raw12 / 16.0f;
    float vcom_v = -((float)(opt[1] / 4)) * 0.1f;

    printf("=== SSD1681 HARDWARE & OTP REGISTER DECODE ===\n");
    printf("0x2F Status/ChipID : 0x%02X -> Chip_ID=%u (%s)\n",
           status, status & 0x03, ((status & 0x03) == 1) ? "Verified SSD1681" : "Unknown IC");
    printf("0x1B Die Temp      : %.2f C (Raw 0x%02X%02X)\n", temp_c, tmp[0], tmp[1]);
    printf("0x2D VCOM Calib    : 0x%02X (%.2f V) | VCOM_OTP_Sel: 0x%02X\n", opt[1], vcom_v,
opt[0]);
    printf("0x2D Mode Matrix   : %02X %02X %02X %02X %02X (Bit=0:Mode1 Full, Bit=1:Mode2
Partial)\n",
           opt[2], opt[3], opt[4], opt[5], opt[6]);
    printf("0x2D Ping-Pong Bit : %s (Byte F[6] = %u)\n",
           (opt[6] & 0x40) ? "ENABLED in OTP" : "DISABLED in OTP (Use SW Sync)", (opt[6] >> 6) &
1);
    printf("0x2D Waveform Ver  : %02X %02X %02X %02X\n", opt[7], opt[8], opt[9], opt[10]);
    printf("0x2E User_ID (10B) : %02X %02X %02X %02X %02X %02X %02X %02X %02X %02X\n",
           uid[0], uid[1], uid[2], uid[3], uid[4], uid[5], uid[6], uid[7], uid[8], uid[9]);
    printf("0x35 Hardware CRC  : 0x%02X%02X\n", crc[0], crc[1]);
}

/* ========================================================================
 * PART 3: Experimental 2-Bit (4-Gray) Synthetic Temperature Override (0x5A / +90C)
 * Note: For laboratory evaluation only. Requires late-revision OTP (+90C slot).
 * ========================================================================== */
void SSD1681_Init_4Gray_OTP_Override(void) {
    EPD_RST_Set(0); EPD_DelayMs(10); EPD_RST_Set(1); EPD_DelayMs(10);
    SSD1681_WaitBusy();
    SSD1681_WriteCmd(0x12); /* SWRESET */
    SSD1681_WaitBusy();

    SSD1681_WriteCmd(0x18); SSD1681_WriteData(0x80); /* Select internal temp sensor */
    SSD1681_WriteCmd(0x22); SSD1681_WriteData(0xB1); /* Read temp & init Mode 1 LUT */
    SSD1681_WriteCmd(0x20); SSD1681_WaitBusy();

    /* Overwrite Temp Register 0x1A with 0x5A0 (+90.0 C) */
    SSD1681_WriteCmd(0x1A);
```

```
    SSD1681_WriteData(0x5A); /* MSB: 0x5A = 90 C */
    SSD1681_WriteData(0x00); /* LSB: 0x00 */

    /* Trigger OTP lookup via 0x91 (loads +90C 4-Gray LUT without re-reading sensor) */
    SSD1681_WriteCmd(0x22); SSD1681_WriteData(0x91);
    SSD1681_WriteCmd(0x20); SSD1681_WaitBusy();
}

static uint8_t Extract_Plane_RAM1(uint8_t b1, uint8_t b2) {
    uint8_t out = 0;
    uint16_t pair = ((uint16_t)b1 << 8) | b2;
    for (uint8_t i = 0; i < 8; i++) {
        uint8_t px = (uint8_t)((pair >> (14 - i * 2)) & 0x03);
        /* 00=White, 01=LightGray, 10=DarkGray, 11=Black */
        if (px == 0x03 || px == 0x01) out |= (uint8_t)(0x80 >> i);
    }
    return ~out; /* Bitwise inverted: White(00)->1, LG(01)->0, DG(10)->1, Black(11)->0 */
}

static uint8_t Extract_Plane_RAM2(uint8_t b1, uint8_t b2) {
    uint8_t out = 0;
    uint16_t pair = ((uint16_t)b1 << 8) | b2;
    for (uint8_t i = 0; i < 8; i++) {
        uint8_t px = (uint8_t)((pair >> (14 - i * 2)) & 0x03);
        if (px == 0x03 || px == 0x02) out |= (uint8_t)(0x80 >> i);
    }
    return ~out; /* Bitwise inverted: White(00)->1, LG(01)->1, DG(10)->0, Black(11)->0 */
}

void SSD1681_Display_4Gray_OTP(const uint8_t *gray2bpp_10000B) {
    SSD1681_WriteCmd(0x24); /* Write Plane 1 to B/W RAM 0x24 */
    for (uint16_t i = 0; i < 10000; i += 2) {
        SSD1681_WriteData(Extract_Plane_RAM1(gray2bpp_10000B[i], gray2bpp_10000B[i + 1]));
    }
    SSD1681_WriteCmd(0x26); /* Write Plane 2 to RED RAM 0x26 */
    for (uint16_t i = 0; i < 10000; i += 2) {
        SSD1681_WriteData(Extract_Plane_RAM2(gray2bpp_10000B[i], gray2bpp_10000B[i + 1]));
    }
    /* Execute Display Mode 1 update via 0xC7 without reloading temperature or LUT */
    SSD1681_WriteCmd(0x22); SSD1681_WriteData(0xC7);
    SSD1681_WriteCmd(0x20); SSD1681_WaitBusy();
}

/* ============================================================================
 * PART 4: Production 1-Bit Full & Partial Refresh Driver (Recommended)
 * ============================================================================ */
static uint8_t g_partial_refresh_count = 0;
#define MAX_PARTIAL_BEFORE_FULL_REFRESH 5

void SSD1681_Init_1Bit_Production(void) {
    EPD_RST_Set(0); EPD_DelayMs(10); EPD_RST_Set(1); EPD_DelayMs(10);
    SSD1681_WaitBusy();
    SSD1681_WriteCmd(0x12); SSD1681_WaitBusy();

    SSD1681_WriteCmd(0x01); SSD1681_WriteData(0xC7); SSD1681_WriteData(0x00);
SSD1681_WriteData(0x00);
    SSD1681_WriteCmd(0x11); SSD1681_WriteData(0x03); /* Standard X+, Y+ top-left (0,0) origin */
    SSD1681_WriteCmd(0x44); SSD1681_WriteData(0x00); SSD1681_WriteData(0x18); /* X: 0..24 (200
px) */
    SSD1681_WriteCmd(0x45); SSD1681_WriteData(0x00); SSD1681_WriteData(0x00);
                            SSD1681_WriteData(0xC7); SSD1681_WriteData(0x00); /* Y: 0..199 (200
px) */
    SSD1681_WriteCmd(0x3C); SSD1681_WriteData(0x05); /* Border follows LUT1 */
```

```
    SSD1681_WriteCmd(0x18); SSD1681_WriteData(0x80); /* Enable real-time internal temp sensor */
    SSD1681_WriteCmd(0x4E); SSD1681_WriteData(0x00);
    SSD1681_WriteCmd(0x4F); SSD1681_WriteData(0x00); SSD1681_WriteData(0x00);
    SSD1681_WaitBusy();
}

void SSD1681_FullUpdate_1Bit_Production(const uint8_t *frame_5000B) {
    SSD1681_Init_1Bit_Production();
    /* Write frame to BOTH 0x24 (NEW) and 0x26 (OLD) so partial refresh is primed */
    SSD1681_WriteCmd(0x24);
    for (uint16_t i = 0; i < 5000; i++) SSD1681_WriteData(frame_5000B[i]);
    SSD1681_WriteCmd(0x26);
    for (uint16_t i = 0; i < 5000; i++) SSD1681_WriteData(frame_5000B[i]);

    SSD1681_WriteCmd(0x22); SSD1681_WriteData(0xF7); /* Temp-compensated Mode 1 Full Update */
    SSD1681_WriteCmd(0x20); SSD1681_WaitBusy();
    g_partial_refresh_count = 0;
}

void SSD1681_PartialUpdate_1Bit_Production(const uint8_t *full_frame_5000B) {
    /* Enforce anti-ghosting full refresh every 5 partial updates (GDEY0154D67.pdf p. 23) */
    if (++g_partial_refresh_count >= MAX_PARTIAL_BEFORE_FULL_REFRESH) {
        SSD1681_FullUpdate_1Bit_Production(full_frame_5000B);
        return;
    }
    EPD_RST_Set(0); EPD_DelayMs(10); EPD_RST_Set(1); EPD_DelayMs(10);
    SSD1681_WaitBusy();

    /* Restore consistent 0x11 = 0x03 (X+, Y+) & lock border 0x3C = 0x80 */
    SSD1681_WriteCmd(0x11); SSD1681_WriteData(0x03);
    SSD1681_WriteCmd(0x3C); SSD1681_WriteData(0x80);
    SSD1681_WriteCmd(0x44); SSD1681_WriteData(0x00); SSD1681_WriteData(0x18);
    SSD1681_WriteCmd(0x45); SSD1681_WriteData(0x00); SSD1681_WriteData(0x00);
                            SSD1681_WriteData(0xC7); SSD1681_WriteData(0x00);
    SSD1681_WriteCmd(0x4E); SSD1681_WriteData(0x00);
    SSD1681_WriteCmd(0x4F); SSD1681_WriteData(0x00); SSD1681_WriteData(0x00);

    SSD1681_WriteCmd(0x24);
    for (uint16_t i = 0; i < 5000; i++) SSD1681_WriteData(full_frame_5000B[i]);

    /* Invoke OTP Display Mode 2 Partial Waveform (0x22 -> 0xFF) */
    SSD1681_WriteCmd(0x22); SSD1681_WriteData(0xFF);
    SSD1681_WriteCmd(0x20); SSD1681_WaitBusy();

    /* Synchronize 0x26 (OLD RAM) for panels without OTP Ping-Pong enabled */
    SSD1681_WriteCmd(0x4E); SSD1681_WriteData(0x00);
    SSD1681_WriteCmd(0x4F); SSD1681_WriteData(0x00); SSD1681_WriteData(0x00);
    SSD1681_WriteCmd(0x26);
    for (uint16_t i = 0; i < 5000; i++) SSD1681_WriteData(full_frame_5000B[i]);
}

void SSD1681_DeepSleep(void) {
    SSD1681_WriteCmd(0x10); SSD1681_WriteData(0x01); /* Deep Sleep Mode 1 (1 uA) */
}
```

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10. Conclusion and Annotated Primary-Source Bibliography

Standardizing on **1-bit bi-tonal operation** (Profile 1 Full Refresh paired with Profile 2 Partial Refresh, a 5-cycle full-refresh anti-ghosting policy, and Deep Sleep shutdown) ensures that firmware operates within

the SSD1681 controller's rated electrical, thermal, and optical specifications across all module revisions from 2019 (`SYX 1920`) through 2025.

- **Solomon Systech Limited (June 2019), *SSD1681: 200 Source x 200 Gate Red/Black/White Active Matrix EPD Display Driver with Controller*, Rev 0.13 (solomon-systech.com):** Primary silicon specification defining the 159-byte Waveform Setting structure (p. 15), 12-bit temperature format (p. 16), 36-slot OTP lookup mechanism (pp. 17–18), and SPI read commands `0x0A`, `0x1B`, `0x2D`, `0x2E`, `0x2F`, and `0x35` (pp. 20–33).
- **LibDriver (2022–2025), *SSD1681 Full-Featured Driver & OTP Verification Reference* (`ssd1681-otp-fingerprinting.md` / `driver_ssd1681.c`):** Documents regional OTP read-back verification via `0x0A` (4 B), `0x2D` (11 B), `0x2E` (10 B), and hardware CRC `0x34`/`0x35` (16-bit), as well as the write-only behavior of LUT register `0x32`.
- **Dalian Good Display Co., Ltd. (Nov 23, 2020), *GDEY0154D67 Product Specifications*, Rev 1.0 (good-display.com/product/388.html):** Module specification documenting the `SSD1681` controller (p. 7), `184 DPI` rating (p. 6), `2.0 s / 1.5 s / 0.26 s` refresh ratings (p. 22), and the 5-operation full-refresh anti-ghosting recommendation (p. 23).
- **Waveshare Electronics (2019–2026), *1.54inch e-Paper V2 Datasheet* (PDF), *Product Catalog* (1.54inch-e-paper.htm), *ESP32 E-Paper & RLCD Tutorial* (docs.waveshare.com), and *Open-Source Driver Repository* (github.com/waveshareteam/e-Paper):** Hardware documentation and open-source driver ecosystem supporting 1.54" V1, V2, (B), and (G) 200×200 displays.
- **Dalian Good Display Co., Ltd. (2021–2025), *GDEY0154D67 Firmware Archives*:** `RP-GDEY0154D67-P-210518.rar` (May 2021), `S-GDEY0154D67-210414.rar` / `-FP-20221011` (Oct 2022), `A32/A8266/AU-GDEY0154D67-FP-20221013.rar` (Oct 2022), and `S/A32-GDEY0154D67-FP4G-20250212.rar` (Feb 2025).